Colours in HCI

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δοκιμος

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To preach the gospel in the regions beyond you, and not to boast in another man's line of things made ready to our hand. But he that glorieth, let him glory in the Lord. For not he that commendeth himself is approved, but whom the Lord commendeth.– 2 Corinthians 10:16-18

Examine yourselves, whether ye be in the faith; prove your ownselves. Know ye not your ownselves, how that Jesus Christ is in you, except ye be reprobates? But I trust that ye shall know that we are not reprobates.– Now I pray to God that ye do no evil; not that we should appear approved, but that ye should do that which is honest, though we be as reprobates.
– 2 Corinthians 13:5-7

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For first of all, when ye come together in the church, I hear that there be divisions among you; and I partly believe it. For there must be also heresies among you, that they which are approved may be made manifest among you.
– 1 Corinthians 11:18-19

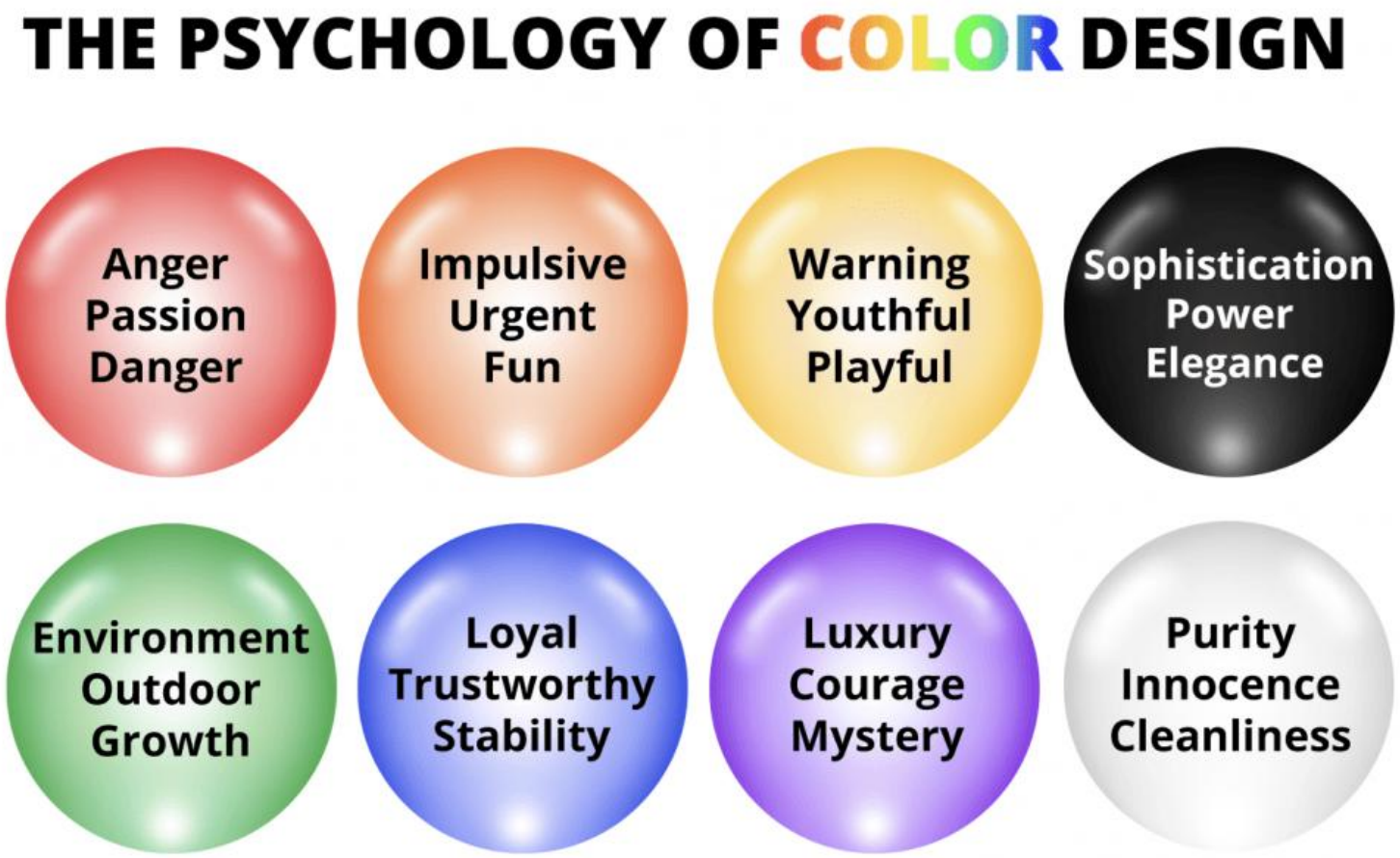
Blessed is the man that endureth temptation: for when he is tried, he shall receive the crown of life, which the Lord hath promised to them that love him.– James 1:12

Colors and Emotions

Color can be used in various ways to enhance the user experience and convey information effectively.

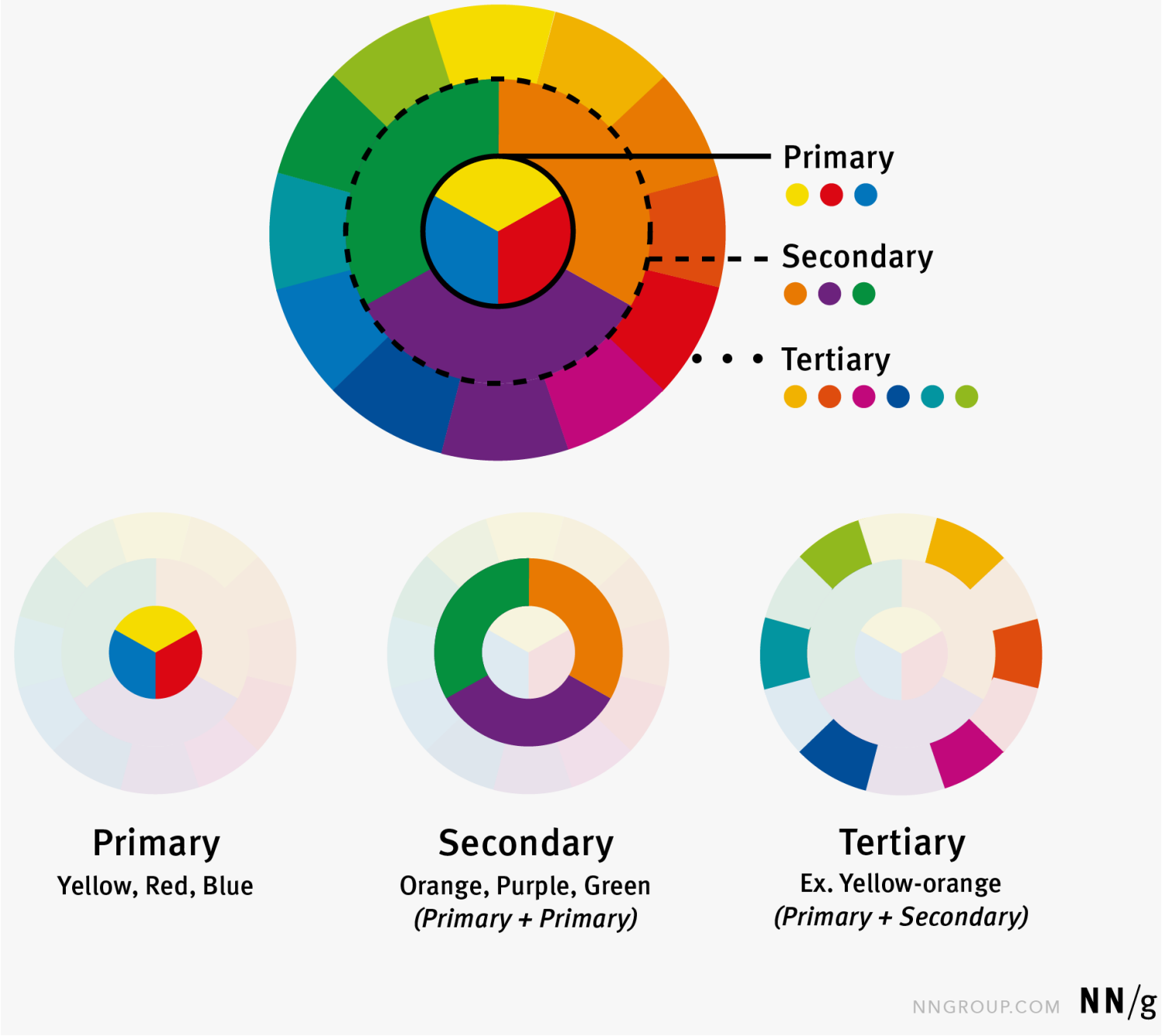
Different colors are associated with specific emotions or concepts. For example;

- blue is often associated with trust,
- red with urgency or danger,
- green with nature or success.



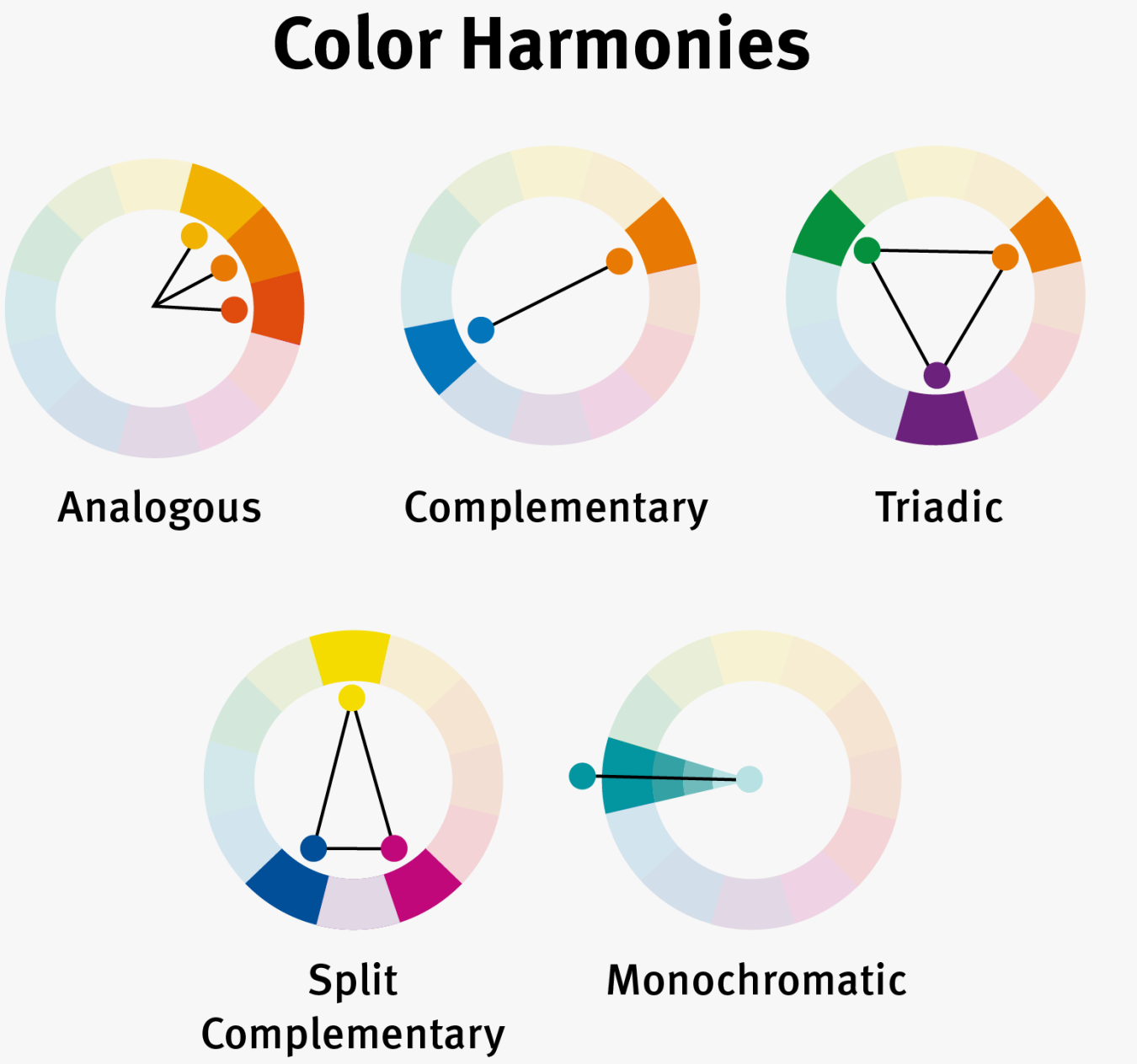
- The color wheel shows 3 colors groups.

- Primary colors are shown in the smallest center circle.
- Secondary colors are in the middle ring
- The largest outer circle is made of tertiary (and all primary and secondary) colors.



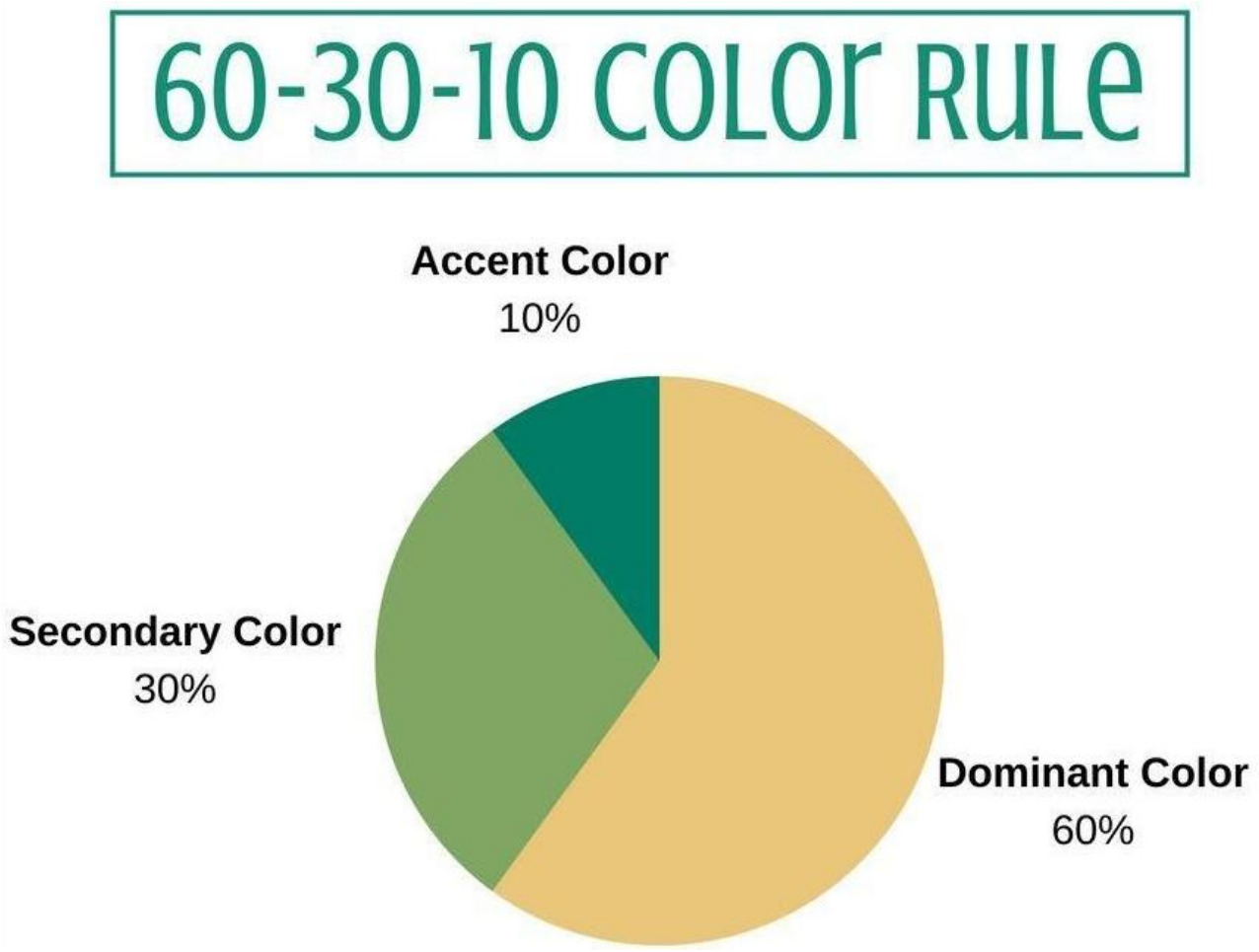
- Color harmonies are the building blocks or the underlying template of a color palette.

- **Analogous:** colors that are next to each other on the color wheel (This color harmony creates low color contrast.)
- **Complementary:** colors that are opposite on the color wheel, which produce high color contrast
- **Split-complementary:** a color combined with others from either side of its complementary color. This harmony slightly softens the contrast from basic complementary colors.
- **Triadic:** three equidistant colors (120 degrees apart) on the color wheel
- **Monochromatic:** tones and shades of a single hue



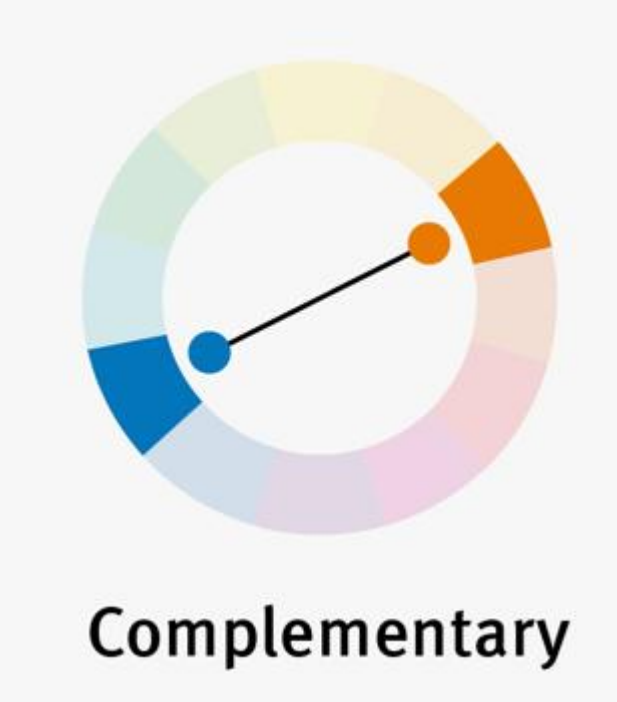
Color Schemes

- Harmonious color schemes can create a pleasing and easy-to-read design.
- Limit your palette to 3 colors.
- Apply 60%- 30% - 10% Rule
- Maintain a consistent use of color throughout your interface to create a cohesive and predictable user experience.



Color Contrast

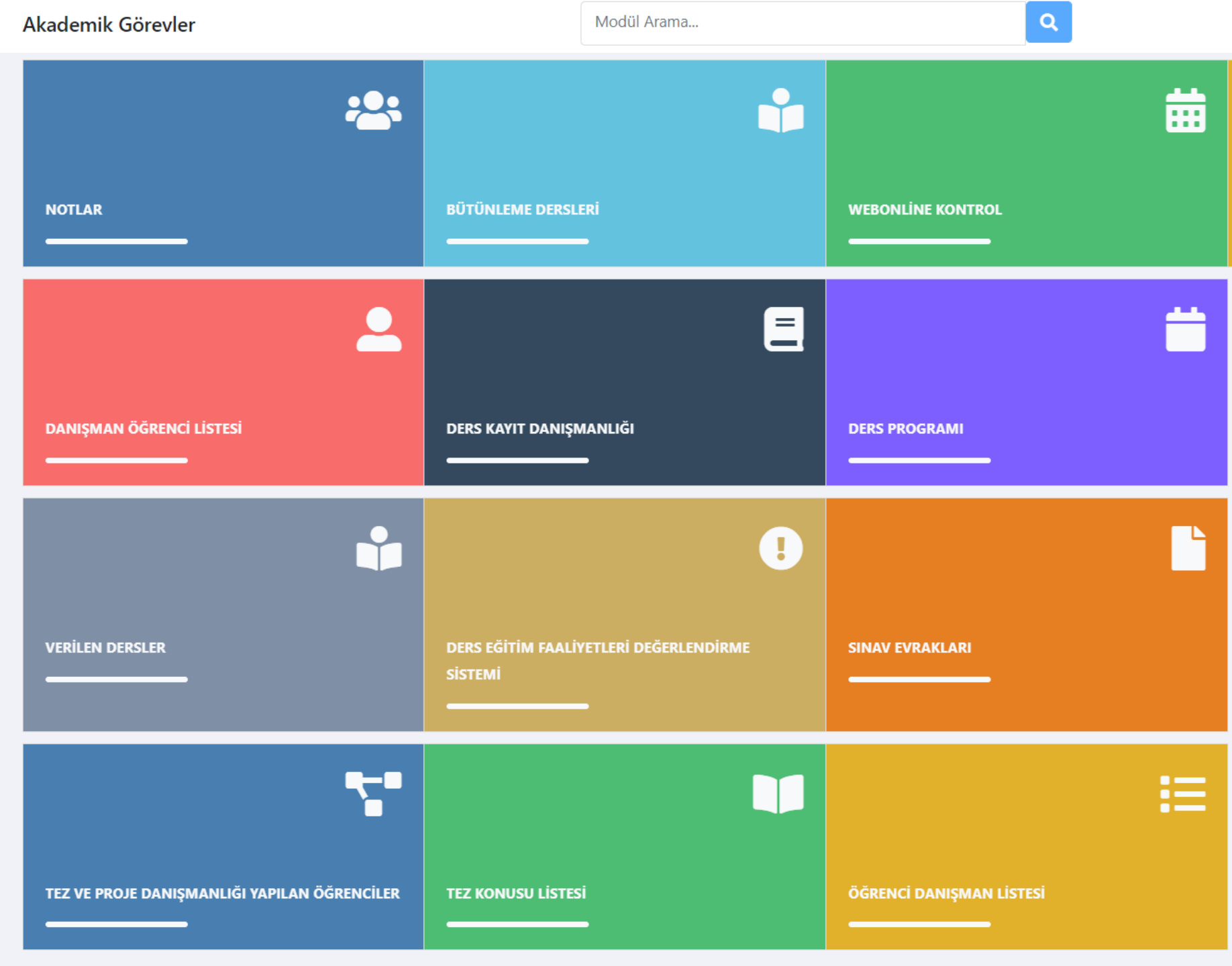
- Ensuring adequate contrast between text and background colors is crucial for readability, especially for users with visual impairments.



Color Coding

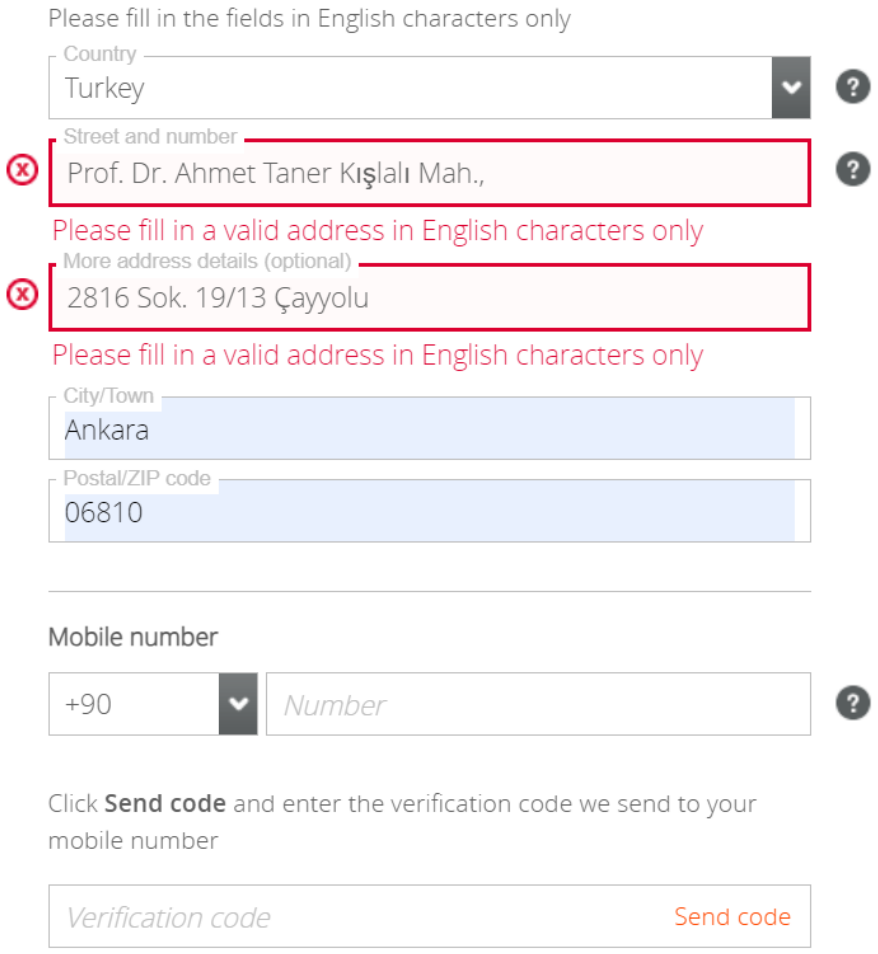
Using color to code or categorize information can help users quickly identify and understand relationships between elements.

For example, different colors for different categories of items in a list or calendar.



Feedback and Validation

Colors can be used to provide feedback to users. For instance, changing the color of a form field border to red when there is an error in the input. Green can be used to indicate success.

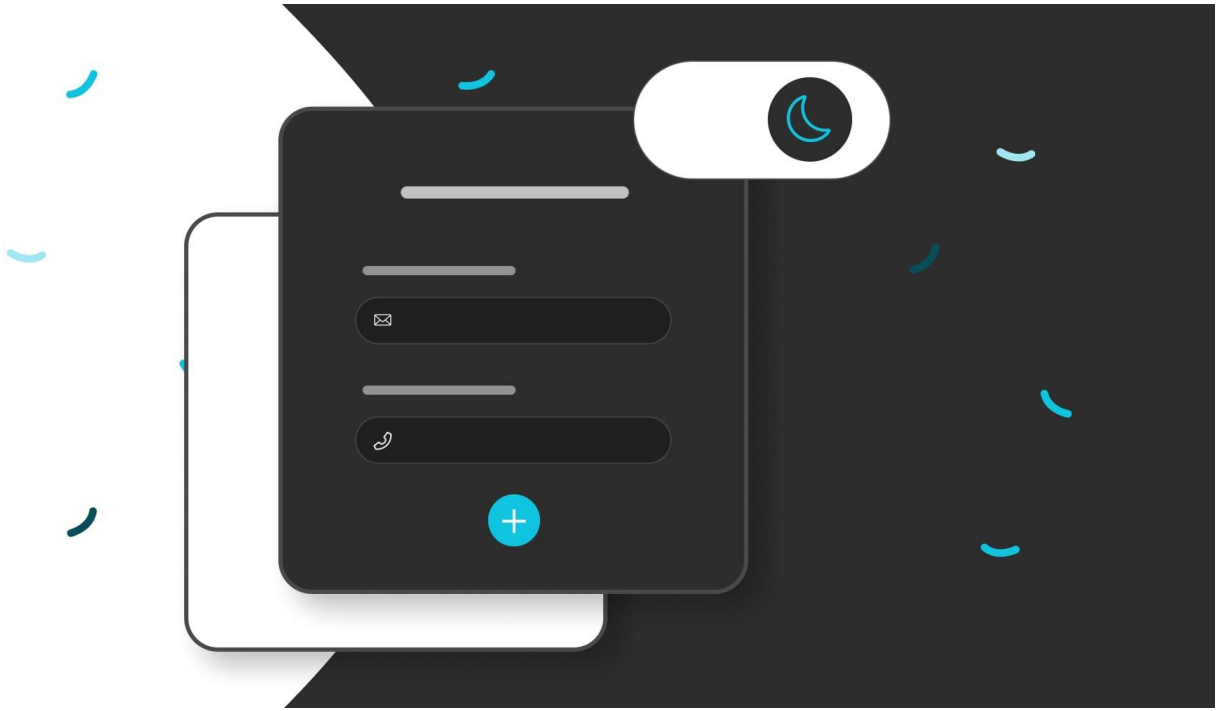


Progress Indicators:

Color can be used in progress bars to indicate the completion of a task. For instance, the bar might turn from red to green as the task progresses.

Dark Mode

Dark mode, which uses darker color schemes, can be more comfortable for users in low-light environments and can help reduce eye strain.



Color Blindness Considerations

It's important to design interfaces with color blindness in mind.

Avoid relying solely on color to convey information, and use patterns, labels, or other cues in conjunction with color.